

1-1c Base of the Pyramid and Reverse Innovation

The global economy can be viewed as a pyramid (Figure 1.2). The top consists of about one billion people with per capita annual income of US\$20,000 or higher. These are mostly people who live in the developed economies in the [Triad \(North America, Western Europe, and Japan.\)](#), which consists of North America, Western Europe, and Japan. Another one billion people earning US\$2,000 to US\$20,000 per year make up the second tier. The vast majority of humanity—about five billion people—earn less than US\$2,000 per year and comprise the [base of the pyramid \(BoP\) \(Economies where people make less than \\$2,000 per capita per year.\)](#)

. Most MNEs focus on the top and second tiers and end up ignoring the BoP markets. An increasing number of such low-income countries have shown a great deal of economic opportunities as income levels have risen. More Western MNEs, such as General Electric (GE), are investing aggressively in the BoP and leveraging their investment to tackle markets in both emerging and developed economies.

Figure 1.2

The Global Economic Pyramid

Sources: Adapted from (1) C. K. Prahalad & S. Hart, 2002, The fortune at the bottom of the pyramid, *Strategy + Business*, 26: 54–67; (2) S. Hart, 2005, *Capitalism at the Crossroads*, Philadelphia: Wharton School Publishing, 111.

One interesting recent development out of emerging economies is [reverse innovation \(An innovation that is adopted first in emerging economies and is then diffused around the world.\)](#)—an innovation that is adopted first in emerging economies and then diffused around the

world. Traditionally, innovations are generated by Triad-based multinationals, with the needs and wants of rich customers at the top of the pyramid in mind. When such multinationals entered lower-income economies, they tended to simplify the product features and lower prices. In other words, the innovation flow is *top-down*. However, as Deere & Company found out in India, its large-horsepower tractors designed for American farmers were a poor fit for the different needs and wants of Indian farmers. Despite Deere's efforts to simplify the product and reduce the price, the price was still too high in India. Instead, Mahindra & Mahindra brought its widely popular small-horsepower tractors that were developed in India to the United States and carved out a growing niche that eventually propelled it to be the world's *largest* tractor maker by units sold. (Mahindra & Mahindra is now so committed to the United States that it sponsors bull-riding tournaments in Texas.) In response, Deere abandoned its US tractor designs and "went native" in India, by launching a local design team charged with developing something from scratch—with the needs and wants of farmers in India (or, more broadly, in emerging economies) in mind. The result was a 35-horsepower tractor that was competitive with Mahindra & Mahindra not only in India, but also in the United States and elsewhere. In both cases, the origin of new innovations is from the BoP. The flow of innovation is bottom-up—in other words, reverse innovation.

Does this Mahindra & Mahindra tractor developed from a BoP market have potential elsewhere?

India Pictures RM/Dinodia Photos/Alamy Limited

The reverse innovation movement suggests that emerging economies are no longer merely low-cost production locations or attractive new markets (hence the term "emerging markets"). They are also sources of new innovations that may not only grow out of BoP markets, but also have the potential to go uphill to penetrate into the top of the global economic pyramid. For example, a Chinese start-up, Xiaomi, has recently dethroned Samsung and Apple in the smartphone market in both China and India, selling its

smartphones for only US\$100 (see [Emerging Markets 1.1](#)). Relative to a feature-rich US\$600 Apple iPhone or a US\$500 Samsung Galaxy, a Xiaomi phone is merely good enough. It is 3G-capable and has a solid processor, a passable camera, and barely decent but expandable memory (8 GB), which can be expanded to 64 GB with cheap SD cards. But its performance is certainly more than 20% of an Apple or a Samsung. Thus, to customers in the BoP and beyond, Xiaomi's reverse innovation delivers tremendous value relative to its price. In a *Harvard Business Review* article, Jeff Immelt, chairman and CEO of a leading practitioner of reverse innovation, GE, noted:

Emerging Markets 1.1

Fighting in and out of the Chinese Smartphone Industry

Commanding one-third of worldwide sales, China is now the largest smartphone market in the world. Not surprisingly, global leaders Samsung and Apple (in that order) sell a lot in China, which absorbs approximately 20% of their output. What is interesting is that six of the top eight vendors are Chinese firms, and that neither Samsung nor Apple is the volume leader. Competing intensely among themselves, the "Gang of Six" consists of computer king Lenovo, telecom equipment giants Huawei and ZTE, consumer electronics firms TCL and Coolpad, and red-hot start-up Xiaomi.

Which firm is the market leader by volume in China? Surprise: it is Xiaomi (pronounced "shee-owl-mee," meaning "Little Rice"). In the second quarter of 2014, Xiaomi, which only sells online, dethroned Samsung to be the market champion by volume, with a 14% market share. Xiaomi's secrets? From a resource-based view, plenty. Fast prototyping, with very short "launch-test-improve" cycles. Offering special software not available in other Android devices. Undercutting rivals with rock-bottom prices—US\$100 for most Xiaomi models vis-à-vis Samsung's high-end Galaxy smartphones that retail for US\$500. Imitating leading brands—Xiaomi's founder, Lei Jun, is famous for wearing Steve Jobs-style black T-shirts and jeans when showing off new models on stage. Overall, Xiaomi grew 240% in the second quarter of 2014, compared with the second quarter of 2013. Its extraordinary performance propelled it to become the fifth-largest smartphone player in the world—behind Samsung, Apple, Huawei, and Lenovo (in that order).

Chinese smartphone makers are naturally salivating about global markets. Xiaomi only sells 3% of its smartphones outside of China, Lenovo 16%, and Huawei 41%. They are likely to drive smartphones' commoditization—a process of competition through which unique products that command high prices and high margins are no longer able to do so, thus becoming commodities. Just as in China, Xiaomi at US\$98 apiece has rapidly become the market leader in India. But here is a catch from an institution-based view. As they increasingly venture outside China, similarities in design between Chinese brands and global leaders—and potential intellectual property (IP) infringement inside the devices—are likely to incur the

wrath of Apple and Samsung. IP disputes, of course, are nothing unusual among smartphone giants. Apple and Samsung themselves fought nasty court battles for years. Xiaomi and other Chinese smartphone makers have armed themselves with Google executives and Silicon Valley lawyers seasoned at navigating the perilous waters between war and peace in IP. While the last page of Chinese smartphone makers' story is not likely to be written any time soon, their performance will ultimately be driven by a combination of their technological and marketing capabilities and their institutional and legal savvy.

Sources: Based on (1) *Bloomberg Businessweek*, 2014, *Samsung's China problems come to India*, October 27: 44–45; (2) *Economist*, 2014, *Smartening up their act*, October 25; (3) *Forbes*, 2014, *China's Xiaomi becomes world's 5th largest smartphone maker*, July 31; (4) T. Hout & D. Michael, 2014, *A Chinese approach to management*, *Harvard Business Review*, September; (5) *South China Morning Post*, 2014, *Chinese companies drive commoditization of smartphone market*, August 18; (6) *Wall Street Journal*, 2014, *Xiaomi overtakes Samsung in China smartphone market*, August 4; (7) P. Williamson & E. Yin, 2014, *Accelerated innovation*, *MIT Sloan Management Review*, summer.

To be honest, the company is also embracing reverse innovation for defensive reasons. If GE doesn't come up with innovations in poor countries and take them global, new competitors from the developing world—like Mindray, Suzlon, Goldwind, and Haier—will ... GE has tremendous respect for traditional rivals like Siemens, Philips, and Rolls-Royce. But it knows how to compete with them; they will never destroy GE. By introducing products that create a new price-performance paradigm, however, the emerging giants very well could. Reverse innovation isn't optional; it is oxygen.

As advised by GE's Immelt, today's students—and tomorrow's business leaders—will ignore the opportunities and challenges at the BoP at their own peril. This book will help ensure that *you* will not ignore these opportunities and challenges.

Chapter 1: Globalizing Business: 1-1c Base of the Pyramid and Reverse Innovation

Book Title: WGU Economics

Printed By: Mike Benton (mbento6@wgu.edu)

© 2021 Cengage Learning, Cengage Learning

© 2026 Cengage Learning Inc. All rights reserved. No part of this work may be reproduced or used in any form or by any means - graphic, electronic, or mechanical, or in any other manner - without the written permission of the copyright holder.